

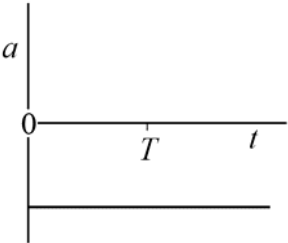
| Question     | Marking guidance                                              | Mark     | Comments                                                                                                                                                                                                                                 |
|--------------|---------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02.1         | (Gravitational) attraction between the weight and the Earth ✓ | 1        |                                                                                                                                                                                                                                          |
| 02.2         | Correct detail about what constitutes action–reaction pairs ✓ | 1        | eg (the two forces) must be of the same type OR co-linear OR in opposite directions OR of equal magnitude OR act on different objects. Any one correct attribute.<br>The second mark is NOT dependent on having a correct answer to 02.1 |
| <b>Total</b> |                                                               | <b>2</b> |                                                                                                                                                                                                                                          |

| Question | Marking guidance                                       | Mark | Comments |
|----------|--------------------------------------------------------|------|----------|
|          | The energy equivalent of the mass of the particle... ✓ |      |          |

|    |   |
|----|---|
| 19 | D |
|----|---|

|    |   |
|----|---|
| 16 | C |
|----|---|

|    |          |                                             |     |
|----|----------|---------------------------------------------|-----|
| 12 | <b>C</b> | a ball falling vertically at terminal speed | AO1 |
|----|----------|---------------------------------------------|-----|

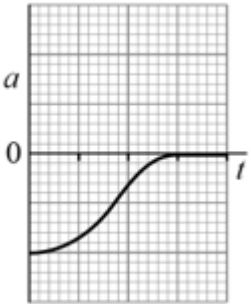
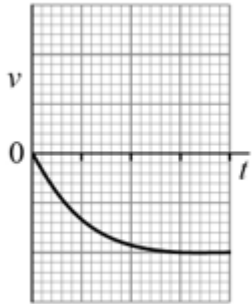
|    |   |                                                                                     |     |
|----|---|-------------------------------------------------------------------------------------|-----|
| 15 | A |  | AO3 |
|----|---|-------------------------------------------------------------------------------------|-----|

|    |          |                                   |     |
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| 17 | <b>A</b> | increasing the weight of the ball | AO2 |
|----|----------|-----------------------------------|-----|

|    |          |                                                                                        |     |
|----|----------|----------------------------------------------------------------------------------------|-----|
| 18 | <b>B</b> | the gravitational force of the Earth on a box and the reaction of the box on the Earth | AO3 |
|----|----------|----------------------------------------------------------------------------------------|-----|

| Question | Answers                | Additional comments/Guidelines               | Mark | AO  |
|----------|------------------------|----------------------------------------------|------|-----|
| 01       | $\text{kg m s}^{-2}$ ✓ | units in any order<br>must be all lower case | 1    | AO1 |
| Total    |                        |                                              | 1    |     |

| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
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|    |   |                                                                                                                                                                                |     |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 15 | C | <div></div> | AO3 |
|----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Additional comments/Guidelines                                                                                                                                                                           | Mark | AO                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------|
| 06       | <p>use of <math>E = \frac{1}{2}mv^2</math> <b>OR</b> use of <math>\Delta E = Fs</math> ✓</p> <p><math>\frac{1}{2}m \times 15^2 - \frac{1}{2}m \times 13^2 = \frac{1}{2}mv^2</math></p> <p><b>OR</b> recognition that energy difference is final energy of <b>Q</b> ✓</p> <p>7.5 (m s<sup>-1</sup>) ✓</p> <p><b>alternative:</b></p> <p>Use of kinematics equations to obtain an expression for <b>a</b> for <b>P</b> ✓</p> <p>(expression for <b>a</b> found and) substituted into <math>v^2 = u^2 + 2as</math> for <b>Q</b> ✓</p> <p>7.5 (m s<sup>-1</sup>) ✓</p> | <p>‘use of’ here means rearrangement and partial substitution</p> <p>Calculator value is 7.483314774</p> <p>In MP1 look for <math>a = (-)\frac{13^2}{2s}</math></p> <p>Condone incorrect sign in MP2</p> | 3    | <p>AO1 × 2</p> <p>AO2 × 1</p> |
| Total    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                          | 3    |                               |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
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| Question | Answers                                                                    | Additional comments/Guidelines | Mark | AO  |
|----------|----------------------------------------------------------------------------|--------------------------------|------|-----|
| 07.1     | Both act on the same object<br><b>OR</b><br>are different types of force ✓ |                                | 1    | AO3 |

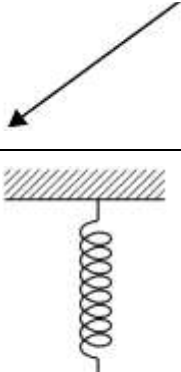
| Question | Answers                                                                                                                                                                                                                                | Additional comments/Guidelines                                                                                                                                                  | Mark | AO                 |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------|
| 07.2     | Drag increases with speed ✓<br><br>Resultant force decreases (since drag increases and thrust is constant) ✓<br><br>When drag = thrust, resultant force is zero (so constant speed) and correct reference to Newton's 1st or 2nd law ✓ | Ignore references to friction<br>Accept air resistance for drag<br>Accept driving force for thrust<br>Condone engine force<br>Accept balanced force for resultant force is zero | 3    | AO1 × 2<br>AO2 × 1 |

| Question     | Answers                                                                                                                                                                                                                   | Additional comments/Guidelines                                                        | Mark     | AO  |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------|-----|
| 07.3         | (Effective surface area decreases) which decreases drag ✓<br>Idea that there is a resultant force acting which causes the car to accelerate<br><b>OR</b><br>Idea that resultant force zero at greater terminal velocity ✓ | Accept air resistance for drag<br>MP2 must follow from correct ideas about the forces | 2        | AO3 |
| <b>Total</b> |                                                                                                                                                                                                                           |                                                                                       | <b>6</b> |     |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
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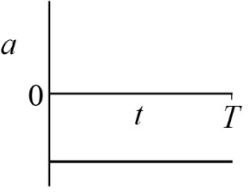
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| 13 | D | $m(a + g) + D$ | AO3 |
|    |   |                |     |

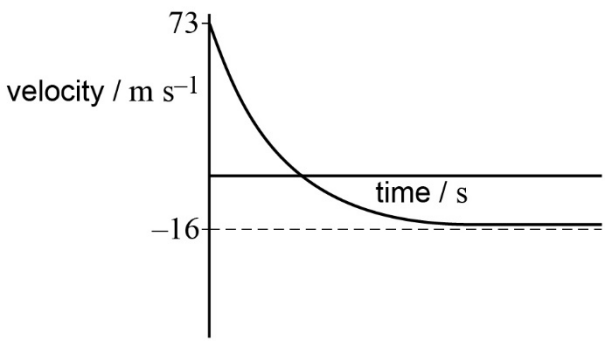
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|----|---|-----------------------------------------------------------------------------------|-----|
| 14 | D |  | AO2 |
|    |   |                                                                                   | AO3 |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Additional comments/Guidelines                                                                                                                  | Mark | AO                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------|
| 03       | <p>Idea that <math>L</math> is (constantly) increasing (as it is a function of speed through the air which is constantly increasing) ✓</p> <p>Idea that <math>C</math> decreases to zero at the moment of take-off (since <math>C + L = W</math> and <math>L</math> is increasing and <math>W</math> is constant) ✓</p> <p>Idea that <math>D</math> is (constantly) increasing as air resistance is a function of speed (through the air) which is (constantly) increasing ✓</p> | <p>Accept reference to increase in drag as flaps are deployed during take-off for MP3</p> <p>Accept reference to vertical component of drag</p> | 3    | <p><math>2 \times \text{AO1}</math></p> <p><math>1 \times \text{AO2}</math></p> |
| Total    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                 | 3    |                                                                                 |

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
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|    |   |                                                                                     |     |
|----|---|-------------------------------------------------------------------------------------|-----|
| 15 | A |  | AO3 |
|----|---|-------------------------------------------------------------------------------------|-----|

| Question | Answers                                                                                                                                                                                                                                                                 | Additional comments/Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mark | AO  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 06.1     | <p>Starts at <math>73 \text{ (m s}^{-1}\text{)}</math> and decreases with correct shape above the <math>x</math>-axis ✓</p> <p>Same continuous shaped curve below the <math>x</math>-axis ✓</p> <p>Becomes constant at <math>-16 \text{ (m s}^{-1}\text{)}</math> ✓</p> |  <p>The graph shows velocity in <math>\text{m s}^{-1}</math> on the vertical axis and time in <math>\text{s}</math> on the horizontal axis. The curve starts at <math>73</math> on the vertical axis, decreases with a concave-up shape, crosses the horizontal axis, and then levels off at <math>-16</math> on the vertical axis. A dashed horizontal line is drawn at <math>-16</math> to indicate the asymptote.</p> | 3    | AO3 |

| Question | Answers                                                                                           | Additional comments/Guidelines | Mark | AO  |
|----------|---------------------------------------------------------------------------------------------------|--------------------------------|------|-----|
| 06.2     | <p>Resolves vertically or uses <math>s = vt</math> ✓</p> <p><math>t = 29 \text{ (s)}</math> ✓</p> |                                | 2    | AO2 |

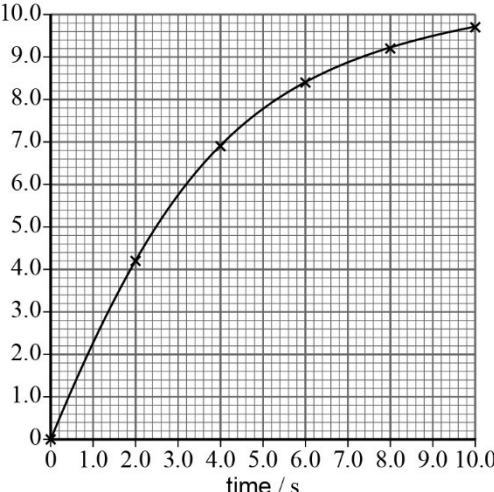
| Question | Answers                                                                                                                                            | Additional comments/Guidelines                                  | Mark | AO                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|-------------------------------|
| 06.3     | Tangent drawn at $t = 5.0 \text{ s}$ ✓<br>Attempts to calculate gradient with a sufficiently large triangle ✓<br>Uses $mg - R = ma$ ✓<br>600 (N) ✓ | expect to see answer in range 2.50–3.00<br><br>allow 575 to 620 | 4    | 1 × AO1<br>2 × AO2<br>1 × AO3 |

| Question | Answers                                                                                                                                                                                                                                                                                                  | Additional comments/Guidelines     | Mark | AO  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------|-----|
| 06.4     | Gradient initially steep as the acceleration is $g$ ✓<br>Due to $mg >$ air resistance / drag or large resultant force ✓<br>Gradient decreases due to drag increasing / resultant force decreasing so acceleration decreases ✓<br>When drag = weight, jumper reaches constant speed so gradient is zero ✓ | Allow air resistance is negligible | 4    | AO3 |

|       |  |  |    |  |
|-------|--|--|----|--|
| Total |  |  | 13 |  |
|-------|--|--|----|--|

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| Question | Answers | Additional comments/Guidelines | Mark | AO |
|----------|---------|--------------------------------|------|----|
|----------|---------|--------------------------------|------|----|

| Question | Answers                              | Additional comments/Guidelines                                                                                                                       | Mark | AO  |
|----------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 11.1     | Appropriate line of best fit drawn ✓ | <div><div>speed / m s<sup>-1</sup></div><div>time / s</div></div> | 1    | AO4 |

| Question | Answers                                                                                                                                                   | Additional comments/Guidelines                     | Mark | AO  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------|-----|
| 11.2     | Tangent to the line drawn on the steepest part of the line ✓<br>Their acceleration $\times 95$ ✓<br>Answer between 300 and 500 (N) when rounded to 1 sf ✓ | Max 1 if their acceleration is not from a gradient | 3    | AO3 |

| Question | Answers                                                                                                                                                                                                                                      | Additional comments/Guidelines                                                                                                                                                                                                                            | Mark | AO  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| 11.3     | Air resistance increases with speed ✓<br><br>Idea that resultant force = force exerted by rider – air resistance, so resultant force decreases ✓<br><br>Gradient is proportional to force / gradient is acceleration so gradient decreases ✓ | Allow drag / air friction for air resistance<br><br>Condone resistive forces / friction in MP2 only<br><br>“Gradient is acceleration” alone is not enough for MP3<br><br>If no other mark awarded, then allow 1 mark for linking gradient to acceleration | 3    | AO3 |

|              |  |  |          |  |
|--------------|--|--|----------|--|
| <b>Total</b> |  |  | <b>7</b> |  |
|--------------|--|--|----------|--|

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| Question | Key | Answer | AO |
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Unknown Unknown - Unit 1 - Q01

| Question | Marking guidance                                                                                                                                                                                               | Mark | Comments                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01.1     | <p>1000 (N) <b>and</b> 6000 (N) seen</p> <p><b>OR</b></p> <p><math>F = \sqrt{(1000)^2 + (6000)^2}</math> ✓ allow incorrect values seen<br/>                     = 6083 (N) ( = 6100) ✓ More than 2 sf seen</p> | 2    | <p>Independent marks</p> <p>Allow full credit for appropriate scale drawing<br/>                     Ignore rounding errors in 3<sup>rd</sup> sig fig.</p>                                                                              |
| 01.2     | <p><math>\tan\theta = 1000 / 6000</math> or correct use of sin or cos ✓</p> <p><math>\theta = 9.5</math> (9.46°) ✓</p>                                                                                         | 2    | <p>Allow range 9.4 – 10.4</p> <p>Use of cos yields 10.4</p> <p>Allow use of 6100</p> <p>Some working required for 2 marks</p> <p>Max 1 mark for correct calculation of vertical angle (range 79.6 – 80.6) some working must be seen</p> |

| Question | Marking guidance | Mark | Comments |
|----------|------------------|------|----------|
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